SSHVisual: Visual Tool to Analyze SSH Data

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Abstract

Network operators use visualizations to detect attacks on their systems, exploring and interacting with data to uncover patterns that would otherwise stay hidden. Operators then use these findings to block attackers and to expand the knowledge base of known threats. Detecting anomalies and uncovering deeper patterns iteratively in data is at best, tedious and at worst, not achievable without data visualizations. Previous works in network visualization analyzed the IP and AS dimensions of SSH logins without much focus on the username aspect of the attackers' attempts, detected anomalous network events only where baseline events were available, and enabled operators to share findings with one another. Little attention has been paid to whether attackers observed at different times represent the same persistent actor. We present an iterative visualization SSHVisual, using login attempts collected on our own SSH-facing systems. SSHVisual analyzes the IP, AS, and username dimensions of SSH attacks, in service of SSH campaign detection. It complements automated detection by allowing operators to find anomalous events without normal baseline events. It correlates SSH attacks with real-world events, and it introduces a similarity metric that quantifies attacker persistence over time. Like previous systems, SSHVisual supports iterative and collaborative sharing of findings between operators. We evaluated SSHVisual using case studies and found that it helps us block attacks and uncover patterns. We identified campaigns in which a set of attempted usernames recurred across unrelated IPs and ASes. The similarity metric surfaced attackers returning across different time periods despite changing network origin. SSHVisual shows that treating usernames as a first-class analytic dimension reveals attacker coordination that IP-centric analysis misses, and that exploratory, operator-driven detection remains valuable.

I Introduction

Computer systems worldwide employ SSH [1] to allow their users to securely connect to them in order to execute commands and do remote administration. A lot of users do a username, password combination (password based authentication) in order to gain entry to the SSH system. Attackers, who are adversarial in nature, can use brute force attack mechanisms [2] to get into SSH systems. Sometimes, they guess what the username, password combinations are and other times, they take advantage of poorly secured systems to build botnets [2] to carry out further attacks [3, 4] or steal resources from the systems, like cryptomining [5, 6].

Given the prevalence of SSH attacks in the IPv4 space [2], we collect SSH login data on our own systems and present SSHVisual. SSHVisual serves two kinds of network operators. Network administrators, who handle day-to-day SSH traffic, need to identify attackers quickly and block them. Security analysts, who study traffic over longer time periods, look for patterns that are not immediately apparent and share what they find across organizations. Consider a single time period: an administrator can isolate the highest-volume attacker and block it immediately, while an analyst examining the same window can iteratively surface a set of low-frequency attackers that attempt an identical sequence of usernames, and group them as a coordinated botnet.

Surveys of network visualization [7, 8, 9, 10, 11] establish the maturity of the field and the gaps we target. Our contributions follow from treating the username as an attacker identity. First, actor-level analysis: we treat the attacker as a first-class actor — including the username identity, not only the source IP or autonomous system (AS) — and we identify coordinated campaigns. Since network data affords no nromal baseline for automated detection [12], this identification is necessarily exploratory (therefore, not antagonistic but complementary to automated detection with nromal baselines) — operators recognize coordination visually rather than specifying it in advance. Second, a similarity metric that quantifies attacker similarity and lets operators track how actors evolve in the username dimension across time periods. Third, correlation of actor behavior with real-world events, which is more tractable through usernames than IPs, given the limitations of IP corroboration. Additionally, following established practices, we allow for iterative operator driven collaboration and labeling of irregularities found in the data.

The rest of the paper is as follows: Section II is Related Work, Section III is Tasks and Requirements, Section IV is Design, Section V is Evaluation and Section VI is Conclusion and Future Work.

II Related Work

A long line of tools [13, 14, 15, 16, 17, 18, 19, 20] render network traffic and security events navigable for operators and analysts with the majority being network flow and traffic oriented. Additionally, a smaller alert- and log-oriented group [21, 22, 23, 24, 25, 26, 27, 28, 29, 30] organizes IDS output and heterogeneous logs of which [23] contributes a visual analysis (of the CICIDS2017 dataset) that isolates at-

tack clusters — including SSH brute-force (SSH-Patator) — through clustering and dimensionality reduction, namely a multi-resolution wavelet-feature analysis.

A Attack Campaigns with Time

Honeypot literature does not promote the username identity as an actor in its own right (for attack campaign detection purposes): it remains an attribute to count, never a unit to track — the unit that, alongside the IP and AS, anchors our analysis. Some papers [31, 32, 33, 34, 35, 36, 37] collect and chart honeypot traffic, with several of them capturing SSH credentials (usernames and passwords). Where credentials are charted — the SCADA SSH honeypot [34], kippo-graph [35], and the container deployment [36] — usernames and passwords appear only as static top-N tallies (kippo-graph's top-10 username and password bars are the archetype), never placed on a time axis even where the same tool charts attack volume over time.The WAN honeypot framework [38] is the closest prior system on the credential axis: it computes an IP-to-IP dictionary-overlap (containment) score to probe whether six same-/24 source IPs share a wordlist — a manual, one-off prototype of credential-based coordination analysis, performed on the IP unit and without a temporal dimension.

B Attacker Similarity with Time

Computed similarity between entities is well established in literature [39, 40, 41, 42]. We do not claim similarity computation itself as novel; the contribution lies in the entity it operates on and how it moves through time. On the entity: these systems cluster internal hosts [39], behavioral profiles [41], IP/port members [42], or internal enterprise users grouped by shared resources — not external attacker identities [40]. None treats the username as an attacker actor. On time: [41, 42] reach into the temporal dimension, but both do not compute the similarity of the attacker population across consecutive windows. We apply similarity based metrics — for example, Jaccard overlap of attackers between successive windows — particularly to the username actor unit, capturing how the attacker population persists or churns over time.

C Limits of automated detection

One premise of our approach is that SSH honeypot traffic does not admit the normal baseline that automated anomaly and campaign detection presuppose. The broader literature already documents the fragility of such detection: Sommer and Paxson [43] note that machine-learning anomaly detection is rarely deployed for intrusion detection given the variability of "normal" the cost of errors, and the difficulty of sound evaluation, while Axelsson [44] shows it is throttled by the base-rate fallacy, where the false-positive rate becomes the limiting factor once intrusions are rare relative to benign traffic. Our setting departs from that one entirely, in the direction Gates and Taylor [12] anticipate when they question whether attacks are rare or a stable normal exists, note that normal traffic may itself be malicious and malicious traffic unremarkable, and observe that detection ultimately devolves to an operator investigating clusters of activity by hand. The consequence is visible in the systems cited above: Situ [39] learns a probabilistic model of each IP's behavior and CLIQUE [45] scores each actor against a baseline of its own expected activity, so both presume a per-actor "normal" that near-uniformly hostile, constantly churning traffic cannot supply. Situ nonetheless concedes that automation alone is insufficient — operators "do not want another automated tool with algorithms they do not trust" — and builds a human-in-the-loop system; we reach the same conclusion from a starker starting point, lacking even the per-actor baseline it assumes, and so treat campaign identification as operator-driven from the outset.

D Actor identity and real-world event correlation

The attacks we observe do not arrive at a steady rate; they rise, fall, and change in character as events unfold in the wider world, so understanding them means lining them up against what is happening outside the network. The link is well documented: Bilge and Dumitras [46] found that the number of attacks against a given vulnerability can jump by up to five orders of magnitude — a hundred-thousand-fold increase — once it is publicly disclosed, and Sabottke et al. [47] showed that future exploitation can be predicted from signals in disclosure channels alone. Allodi and Massacci [48] add that a vulnerability's official severity score is a poor guide to whether it is actually attacked, so the dependable way to learn what attackers are reacting to is to watch real activity and line it up against real events, not to read it off a score. We want to do this at the level of a campaign — a group of attackers acting together — rather than one attacker at a time, and that raises a practical problem: a campaign rarely shows up as a single source address. IP addresses do not stay tied to one machine — one host is assigned many addresses over time [49], hosts cycle through new addresses [50], and counting by address can overstate the number of real machines by orders of magnitude [51] — so it is likely that a view organized by address scatters one campaign across dozens of entries that look unrelated. The username pulls those pieces back together: the same login names likely recur across the changing addresses, so grouping by username re-assembles the campaign that an address-by-address view splits apart. A username is not a perfect identity either — names like "root" are tried by almost everyone, and one campaign runs through a whole list of names — so we treat it as one analysis dimension alongside the address and the network it belongs to (its AS), resolving the overlaps with the similarity and grouping methods above.

E Operator labeling and collaboration

Since detection on our data is operator-driven, the findings an operator produces by hand must be captured and shared rather than discarded. In two lines of work [52, 19], interactive labeling has been used to fold the operator's judgments back into

the system. Collaborative analysis has been supported through shared artifacts with [17] and [20] being notable examples. These systems establish that labeling and collaboration are valuable, which is why our labeling-and-sharing mechanism is the step that turns an identified set of usernames into a durable, shared campaign.

III Tasks and Requirements

The design of SSHVisual draws on the authors' experience as network operators and on gaps we identified in prior work on SSH attack analysis and security visualization. From these we derive three tasks and five requirements.

A Tasks

T1: Identify SSH campaigns. Security analysts need to find groups of attackers that behave alike across usernames, IP addresses, autonomous systems (ASes), and countries, and to record those groups so they can be referenced later.

T2: Detect anomalies that warrant action. Network administrators need to notice unusual activity, such as a sudden spike or a single source dominating traffic, quickly enough to decide whether to block it.

T3: Share findings across operators. Analysts and administrators on the same network need to see what their colleagues have already investigated, so that the same patterns are not identified twice.

B Requirements

R1: Attacker similarity profiling. The system should be able to algorithmically cluster similar attackers based primarily along the username dimension, secondarily along IP and AS information and tertiarily along other dimensions. R1 mostly serves T1 as it allows analysts to build a repository of similar attackers.

R2: Sorting and ranking SSH attack data is heavy-tailed: a few usernames [53] and IP addresses [54] account for most attempts, and the remaining entities are buried beneath them. The system should therefore list every entity in each dimension, not only the most active ones, and let the operator sort and rank entities by any combination of attributes. R2 supports T1 and T2: administrators can surface anomalous entities that never reach the top of a volume ranking, and analysts can find entities with similar attribute values.

R3: Shared tagging. Operators should be able to tag attackers, record both what a tag means and why it was applied to a particular attacker, and have those tags persist and remain visible to every operator using the system. Each tag application should record who applied it and when, so that later operators can judge whether a finding is still current. R3 primarily supports T3, and it supports T1 because tags accumulate into a repository of related attackers.

R4: Complementary tabular and temporal views. No single visualization serves every need. A table supports precise lookup and ranking but discards temporal structure; a linechart preserves temporal structure, showing spikes and campaigns at a glance, but conveys values less precisely. The system should provide both views and let operators move between them. An administrator may find and act on an anomaly using the table alone, whereas an analyst may use the table to assemble a set of related attackers and then examine their behavior over time in the linecharts. R4 supports T1 and T2.

R5: Interactive exploration of the linecharts. The linecharts should follow Shneiderman's Visual Information-Seeking Mantra: overview first, zoom and filter, then details-on-demand [55]. They should open with an overview of the full time range, let the operator zoom into a time window and filter which entities are displayed, and reveal details about an individual entity on request. R5 supports T1 and T2: zooming isolates the time window of a suspected campaign, and filtering removes dominant entities that would otherwise mask smaller ones. Additionally, line charts become harder to read accurately as more series are added [56]. Each linechart should therefore show at most ten entities.

IV Design

SSHVisual combines two visual components, a discovery table and a set of linecharts, with two non-visual components, similarity search and shared tagging. We describe the visual design in Section A and the non-visual design in Section B. All figures show SSH login attempts collected between 1 November 2022 and 8 January 2023, a period of 69 days.

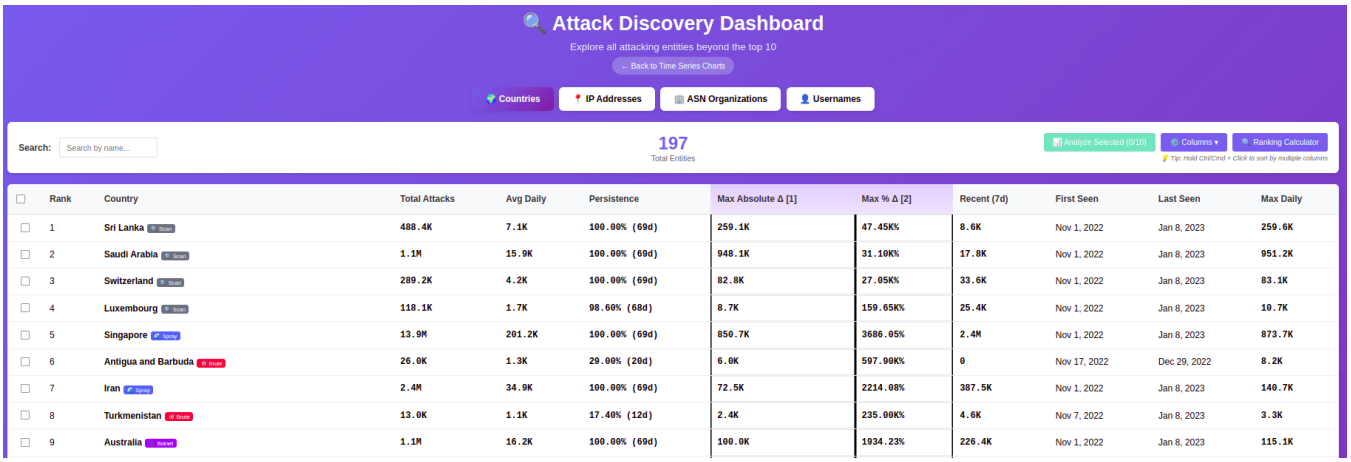
A Visual Design

A.1 Discovery Table

The interface provides two views of the data. The discovery table (Figure 1) is independent of time: each row summarizes one entity over the selected period. The linecharts (Figure 2) can be time-dependent. Both views cover the full data period by default.

The discovery table has four tabs, one per dimension: countries, IP addresses, ASN organizations, and usernames. In Figure 1 the countries tab is active. Most columns are shared across tabs: total attacks, average daily attacks, persistence (the share of days on which the entity was active), the largest absolute and percentage day-over-day change, attacks in the most recent seven days, first- and last-seen dates, and the maximum daily count. Some columns apply only to particular tabs; for instance, the countries tab offers a Unique ASNs column that the ASN tab lacks. Operators choose which columns to display through the Columns menu. Each entity also carries a badge summarizing its dominant behavior, namely Scan, Spray, Botnet or Brute **[describe how badges are assigned]**. A search box filters rows by name, and the counter above the table reports the number of entities in the current tab (197 countries in Figure 1).

Unlike the linecharts, which show only the ten most active entities per dimension, the table lists every entity. Operators can select up to ten rows with the checkboxes and use Analyze



Rank	Country	Total Attacks	Avg Daily	Persistence	Max Absolute Δ [1]	Max % Δ [2]	Recent (7d)	First Seen	Last Seen	Max Daily
1	Sri Lanka	488.4K	7.1K	100.00% (69d)	259.1K	47.45K%	8.6K	Nov 1, 2022	Jan 8, 2023	259.6K
2	Saudi Arabia	1.1M	15.9K	100.00% (69d)	948.1K	31.10K%	17.8K	Nov 1, 2022	Jan 8, 2023	951.2K
3	Switzerland	289.2K	4.2K	100.00% (69d)	82.8K	27.95K%	33.6K	Nov 1, 2022	Jan 8, 2023	83.1K
4	Luxembourg	118.1K	1.7K	98.60% (68d)	8.7K	159.65K%	25.4K	Nov 1, 2022	Jan 8, 2023	10.7K
5	Singapore	13.9M	201.2K	100.00% (69d)	850.7K	3686.05%	2.4M	Nov 1, 2022	Jan 8, 2023	873.7K
6	Antigua and Barbuda	26.0K	1.3K	29.00% (20d)	6.0K	597.90K%	0	Nov 17, 2022	Dec 29, 2022	8.2K
7	Iran	2.4M	34.9K	100.00% (69d)	72.5K	2214.08%	387.5K	Nov 1, 2022	Jan 8, 2023	140.7K
8	Turkmenistan	13.0K	1.1K	17.40% (12d)	2.4K	235.00K%	4.6K	Nov 7, 2022	Jan 8, 2023	3.3K
9	Australia	1.1M	16.2K	100.00% (69d)	100.0K	1934.23%	226.4K	Nov 1, 2022	Jan 8, 2023	115.1K

Figure 1: The discovery table, showing the countries tab. Each row summarizes one entity over the selected period, and operators can sort by any combination of columns or select up to ten rows to plot in the linecharts.

Selected to plot those entities in the linecharts, and they can return to the linecharts at any time. This link lets operators move from a precise, ranked lookup to the temporal behavior of the same entities, fulfilling R4.

A.2 Linecharts

The linechart view (Figure 2) stacks five charts on a shared daily time axis. The first, Total Attacks Over Time, shows the aggregate volume of attempts; the remaining four show the ten most active countries, ASN organizations, IP addresses, and usernames. Each legend entry lists the entity's total for the selected period.

Interaction with the linecharts follows Shneiderman's mantra [55] (R5), and the charts are coordinated so that an action in any one of them updates all of them.

Overview first. On opening, the charts cover all 69 days of data, with each per-entity chart showing the ten most active entities in its dimension.

Zoom. Every chart supports brushing. When the operator brushes a time range on any chart, the range is applied to all five charts and recorded in the Active Filters bar, so the temporal context stays the same whichever dimension the operator is examining. Brushing is disabled once the range is two days or fewer, since a line needs more points to show a trend.

Filter. Left-clicking a legend entry selects that entity and filters every chart to the attacks associated with it. Because each attack attempt links a username, an IP address, an AS, and a country, the selection propagates along these relationships:

- Selecting a country restricts the AS, IP, and username charts to the entities observed in attacks from that country.
- Selecting an AS shows the country it originates from, along with its IP addresses and the usernames they try.
- Selecting an IP address shows its country and AS, along with the usernames it tries.
- Selecting a username shows the countries, ASes, and IP addresses from which it was tried.

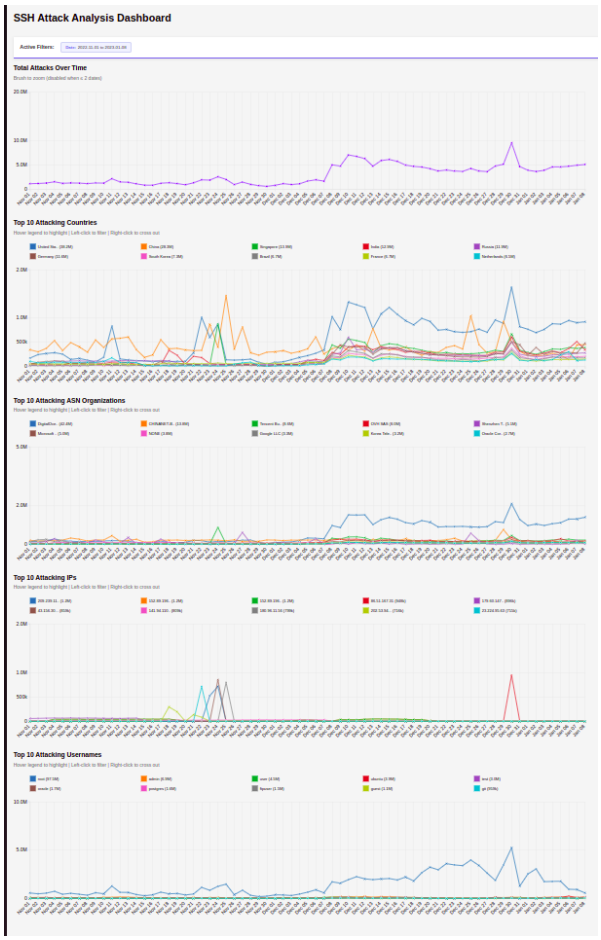


Figure 2: The linechart view. The aggregate chart at the top supports brushing to zoom; the four charts below show the ten most active countries, ASN organizations, IP addresses, and usernames.

Rank	Username	Total Attacks	Avg Daily	Persistence	Max Absolute Δ [1]	Max % Δ [2]	Recent (7d)	First Seen	Last Seen	Max Daily
1	karaf	77.6K	5.2K	21.70% (15d)	32.4K	360.52K%	120	Nov 2, 2022	Jan 8, 2023	44.8K
2	ossuser	82.6K	2.6K	46.40% (32d)	30.4K	276.15K%	690	Nov 3, 2022	Jan 8, 2023	46.0K
3	snow	248.2K	5.2K	69.60% (48d)	22.0K	168.87K%	41.8K	Nov 1, 2022	Jan 8, 2023	35.2K
4	nmrsu	89.4K	1.8K	72.50% (50d)	35.0K	145.91K%	2.9K	Nov 5, 2022	Jan 8, 2023	43.8K
5	lenovo	25.3K	476.51	76.80% (53d)	8.5K	283.10K%	5.7K	Nov 1, 2022	Jan 8, 2023	8.5K
6	fb	9.7K	225.67	62.30% (43d)	4.7K	470.10K%	403	Nov 5, 2022	Jan 8, 2023	4.7K
7	inspur	40.4K	684.71	85.50% (59d)	5.3K	263.05K%	11.2K	Nov 1, 2022	Jan 8, 2023	5.3K
8	radware	12.3K	686.11	26.10% (18d)	4.8K	241.25K%	1.9K	Nov 1, 2022	Jan 8, 2023	4.8K

Figure 3: The usernames tab sorted by Max Absolute Δ (primary key) and Max % Δ (secondary key), surfacing bursty, usernames that may never appear among the top ten by volume.

Right-clicking a legend entry crosses the entity out instead, removing it from its chart and rescaling the y-axis so that smaller series are no longer flattened against the axis by a dominant one.

Details-on-demand. Hovering over a data point shows a tooltip with that entity's attack count for the day, hovering over a legend entry highlights the entity's line, and the legend reports each entity's total for the selected period.

Linked filtering turns the linecharts from four separate rankings into a means of tracing an entity across dimensions. An operator investigating the spike on 30 December, for example, can select `root` in the username chart; the IP, AS, and country charts then show which addresses, networks, and countries tried `root`, and brushing around 30 December narrows the view to the spike itself. Conversely, selecting an AS such as DigitalOcean reveals which usernames its hosts try, indicating whether traffic from a hosting provider reflects one campaign or many. Brushing on every chart, linked filtering, and on-demand details together fulfill R5.

The linecharts answer questions that the discovery table cannot, such as "Do the countries in Figure 1 share the same peaks and valleys?" or "Does one country begin attacking earlier than another?" In Figure 2, the aggregate chart shows a sharp spike on 30 December, and most of the top ten countries peak on the same day, indicating a coordinated, multi-country event rather than activity from a single source. The charts also reveal differences in timing. China's largest peak falls in late November, weeks before the United States' activity rises, and the IP chart shows individual addresses that each spike for a day or two and then fall silent. Neither pattern can be recovered from the table's per-entity totals. Together, Figures 1 and 2 show how the two views complement each other, fulfilling R4.

A.3 Sorting and Ranking

Figure 3 shows the usernames tab sorted on two columns. Operators add sort keys by holding Ctrl/Cmd while clicking column headers, and the bracketed indices in the headers show when the column was clicked.

Sorting matters because the data is heavy-tailed. The username linechart in Figure 2 is dominated by `root`, which accounts for tens of millions of attempts and flattens every other line. Ranking by day-over-day change instead surfaces usernames such as `karaf`, `ossuser`, `lenovo`, `inspur`, and `radware`, several of which are default accounts for specific software or hardware products (`karaf`, for example, is the default account of Apache Karaf). Their totals are far below the top ten, but their activity is concentrated: `karaf` received 77.6K attempts in total, 44.8K of them on a single day, and was active on only 15 of the 69 days. Bursts of product-specific credentials like these suggest campaigns targeting particular devices, yet they are invisible in a volume-ranked chart. Figure 3 thus shows how sorting and ranking over the full entity list fulfills R2.

B Non-Visual Design

B.1 Username Similarity

Similarity search lets operators start from one attacker and retrieve the attackers that behave most like it. For usernames, the unit of analysis is the population of sources that tries a given username. For each username we compute the breadth of that population (the number of distinct IP addresses, ASes, and countries that try it), its volume, its persistence, and how concentrated its traffic is on its single most active source. We do not reuse the seventeen-dimensional profile we apply to IP addresses (Section B.2), because a diagnostic analysis showed several of those features to be uninformative for usernames. Day-to-day stability is the clearest example: most IP addresses are active on many days, so their stability is well defined, but about half of the usernames appear on only a single day, where stability is meaningless. We cluster the username profiles with k-means, selecting $k = 9$. Each cluster is labeled automatically from its centroid. To answer a query, the system retrieves the query's cluster and ranks the other members by Euclidean distance to the query, reported as a similarity score.

In Figure 4, an operator searches for the username `james`. The system places it in Cluster 3, labeled High-Volume Persistent Botnet-Scale, which contains 2,624 usernames; the cluster summary reports that a typical member receives about 31.8K attempts from roughly 4,000 IP addresses and is active on 72% of days. The results table lists the 20 most similar usernames, each with its total attacks, top source country, top three source IPs, and a set of "Why similar?" labels, such as a shared lead source country or comparable volume, that explain the match.

Above the results, a shared infrastructure panel lists IP addresses that appear among the top sources of more than one result, and these addresses are highlighted in the table. In Figure 4, ten addresses are shared. Two of them, 152.89.196.123

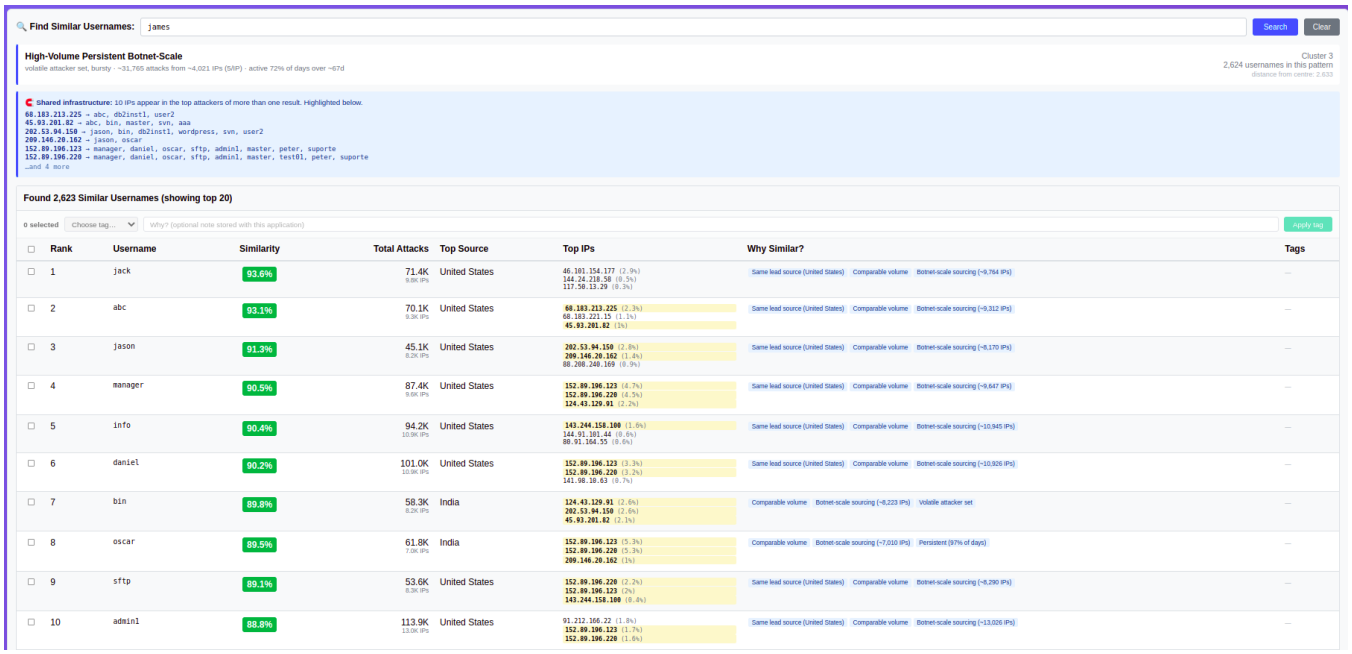


Figure 4: Similarity search for the username `james`. The cluster summary, shared infrastructure panel, and "Why similar?" labels explain why the results were grouped with the query.

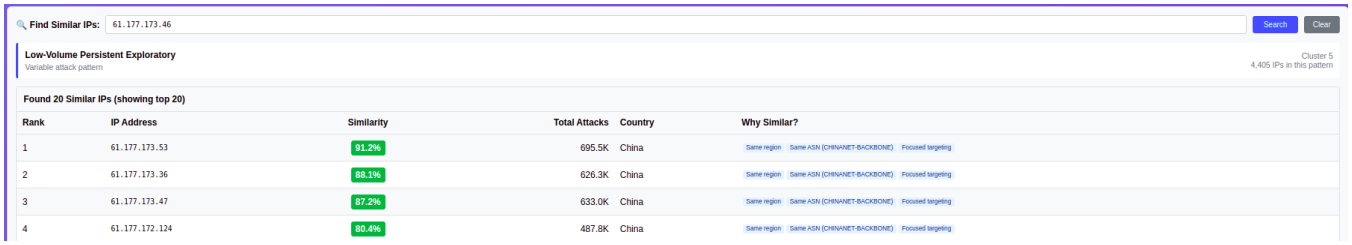


Figure 5: Similarity search for the IP address 61.177.173.46. The nearest neighbors are adjacent addresses on the same network

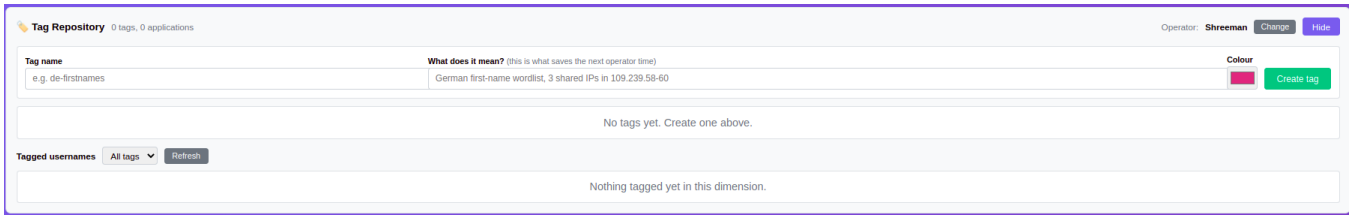


Figure 6: The empty tag repository for the username dimension.

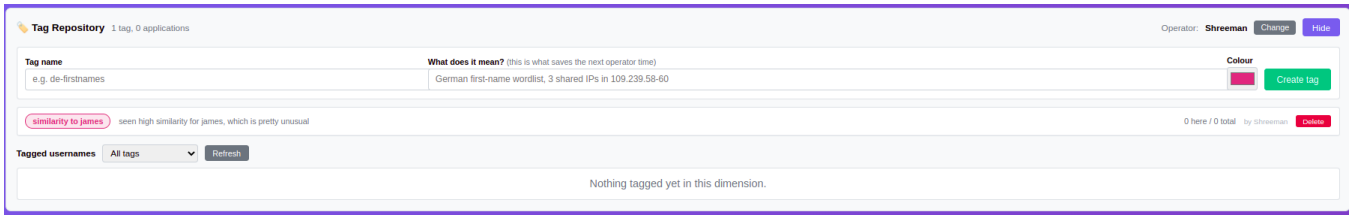


Figure 7: A newly created tag, listed with its description, author, and application counts.

and 152.89.196.220, lie in the same /24 network and are top sources for `manager`, `daniel`, `oscar`, `sftp`, `admin1`, and several other usernames. The panel turns behavioral resemblance into infrastructural evidence: usernames tried by the same hosts are likely drawn from the same credential list and belong to the same campaign.

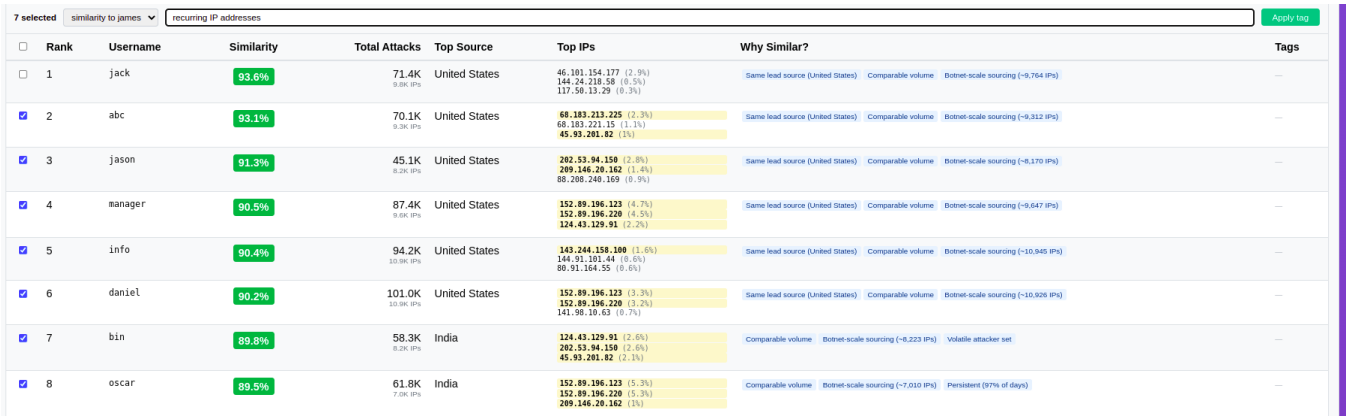


Figure 8: Applying the tag to seven usernames from the similarity results, with a note explaining the application.

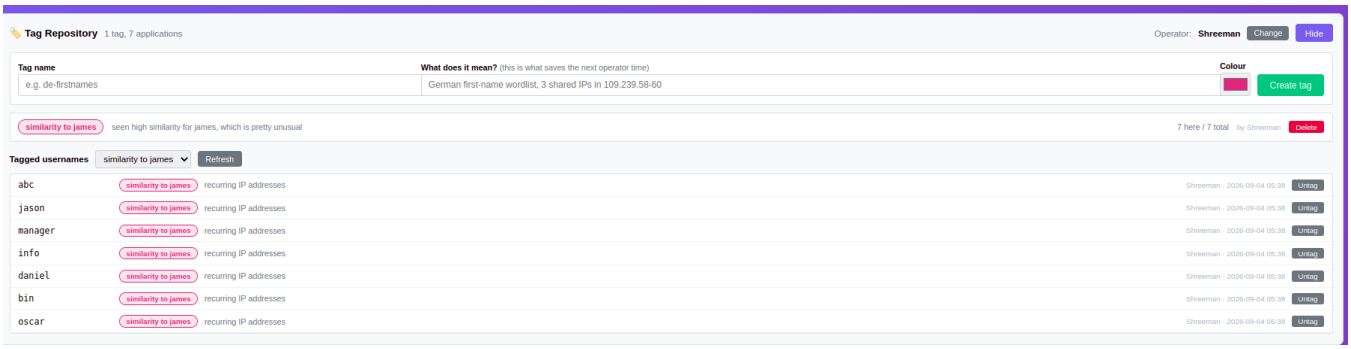


Figure 9: The tag repository after the tag is applied, listing each tagged username with its note, author, and timestamp.

B.2 IP Address Similarity

We characterize each IP address by a seventeen-dimensional behavioral profile drawn entirely from its observed activity during the data period. The features fall into four groups: volume (log-scaled totals, daily means, and peaks); temporal shape (the fraction of days on which the address was active, its burst intensity, its largest day-over-day swings, a fitted trend slope, its activity span, and the share of its traffic in the final week); targeting behavior (the number of distinct usernames it tries, how stable that set is from one day to the next, its rotation rate, and the share of attempts taken by its most-tried username); and two flags indicating whether the address belongs to a cloud provider and whether it originates from a major source country. We cluster these profiles with k-means, label each cluster from its centroid, and answer queries in the same way as for usernames.

In Figure 5, an operator searches for 61.177.173.46, which falls in Cluster 5, labeled Low-Volume Persistent Exploratory, containing 4,405 addresses. The most similar results are 61.177.173.53, 61.177.173.36, 61.177.173.47, and 61.177.172.124, with similarity scores between 91.2% and 80.4%, all belonging to CHINANET-BACKBONE. Behavioral features alone therefore recover infrastructure relationships, suggesting that these hosts are operated together.

Figures 4 and 5 show how SSHVisual lets operators assemble sets of similar attackers along both the username and IP dimensions and see why those attackers were grouped, fulfilling R1.

B.3 Shared Tagging

Figures 6–9 show how an operator records a finding so that other operators can see work that has already been done. Tags are organized in a tag repository for each dimension. The repository displays the current operator's name, and every tag and tag application records its author.

In Figure 6 the repository is empty. To create a tag, the operator gives it a name, describes what it means, and picks a color; the interface prompts for the description with the reminder that it is what saves the next operator time. In Figure 7 the operator has created a tag, "similarity to james," for the username cluster found in Figure 4. The repository now lists the tag with its description, its author, and two counts: applications in the current dimension and applications across all dimensions, both zero so far.

Tags are applied from the similarity results. In Figure 8 the operator selects seven usernames (`abc`, `jason`, `manager`, `info`, `daniel`, `bin`, and `oscar`), chooses the new tag, and adds a note explaining this particular application: "recurring IP addresses." The selection reflects the evidence from the shared infrastructure panel. All seven usernames have highlighted, shared IP addresses among their top sources, whereas `jack`, despite the highest similarity score, has none and is



Figure 10: IPs that behave like 200.195.169.59

left untagged. In Figure 9 the repository reports seven applications. Filtering the list by tag shows each tagged username alongside its note, the operator who applied it, and a timestamp, and any application can be removed with Untag. Tagged usernames also appear in the Tags column of subsequent similarity searches, so an operator who later retrieves the same usernames sees immediately that they have already been investigated.

Separating a tag's meaning from the rationale for each application lets a later operator understand both what a group represents and why each member belongs to it, without repeating the analysis. The recorded author and timestamp indicate whether a finding is recent. Figures 6–9 thus show how tagging supports collaboration among operators, fulfilling R3.

V Evaluation

We evaluate SSHVisual on its ability to uncover patterns. We show this in two ways: 1. finding entities to block that act alike but would be buried underneath entities that dominate attacks and 2. finding campaigns in which the same list of attempted usernames recurs across IPs in different countries. Both the subsections start with a network operator coming across a single IP, for example from an intrusion detection alert, an abuse report, a flag report on stackoverflow or the discovery tables in SSHVisual. On its own, such an IP is one address among tens of thousands; SSHVisual helps the operator find what else it is connected to.

A Finding hidden entities to block

Figure 10 shows the IP similarity search for 200.195.169.59, an IP from Ligga Telecomunicações S.A. in Brazil with 36.4K attacks. The five closest matches (88.6% to 94.9% behavioural similarity) come from Brazil, Indonesia, Italy, Mexico and Australia, each from a different ASN (Not shown in Figure 10 but the top 20 similarity matches – at least 86% – to this IP come from 15 different countries and 21 different ASNs). Despite this spread, they behave alike: each made roughly 30K to 41K attacks, is tagged as exploratory targeting (that is, they use many different usernames), and shares between 1,684 and 1,899 usernames with the searched IP, 46 to 140 of which are uncommon (tried by fewer than 1% of all IPs). The searched IP itself tried 3,821 usernames, 925 of them uncommon. This is the pattern: low-volume IPs working through very large and similarly uncommon username lists from many countries at once.

Figure 11 shows how buried these IPs are despite their interesting nature to administrators. None of the 10 IPs (selected from Figure 10) would appear in the top volume IPs chart. The searched IP ranks #296 of 86,985 IPs and accounts for 0.018% of 206.6M attacks. Its country, Brazil, is large enough to appear in the top 10 countries (#8), but its ASN is 13 times too small to appear in the top 10 ASNs, and the IP itself is 20 times too small to appear in the top 10 IPs. The other IPs rank be-



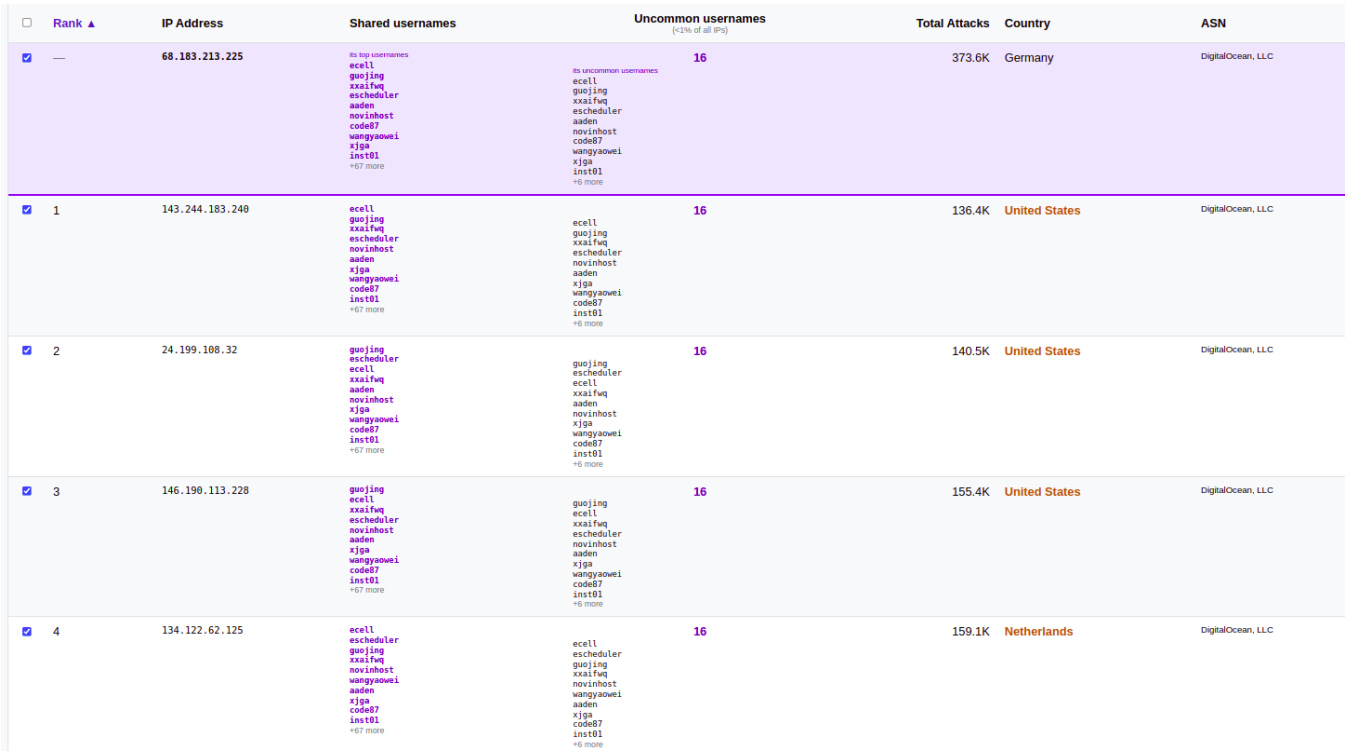
Figure 11: How buried 200.195.169.59 and its similar IPs are

tween #222 and #2,826. An administrator looking only at the high volume charts would see Brazil but never these IPs. With SSHVisual, administrators know not just the IP they found to block but also similar IPs that they should block, and most importantly, why they should block them.

B Username campaigns

Figure 12 shows the username search for IP 68.183.213.225 (DigitalOcean, Germany). Four IPs score 100% username similarity with it: 143.244.183.240, 24.199.108.32 and 146.190.113.228 in the United States, and 134.122.62.125 in the Netherlands (not shown in Figure 12 but at least 12 IPs have a username similarity score greater than 84%). All five try the same 77 usernames in near-identical proportions, and all 16 of the searched IP's uncommon usernames (such as ecell, guojing and xxaifwq) appear in every list. The uncommon usernames matter here: a 100% score built on common usernames like root and admin would mean little, but 16 shared uncommon usernames are unlikely to occur by chance. The five IPs sit on one ASN across three countries (not shown in Figure 12 but the 12 IPs mentioned above come from 5 countries and 6 ASNs), which is consistent with a single operator renting servers in several regions.

Figure 13 shows why this campaign is hard to see from attack volume alone. In the IP chart, each of these IPs is active for only a day or two, and at different times: 146.190.113.228 on Dec 19, 143.244.183.240 on Dec 20, 24.199.108.32 on Dec 28, 68.183.213.225 on Jan 2 and Jan 8, and 134.122.62.125 on Jan 5 to 6. The lines never rise together, so the chart suggests five unrelated bursts. Only the shared username list links them into one campaign.



Rank ▲	IP Address	Shared usernames	Uncommon usernames (<1% of all IPs)	Total Attacks	Country	ASN
—	68.183.213.225	its top usernames: ecell, guojing, xxaifwq, escheduler, aaden, novinhost, code87, wangyaowei, xjga, inst01 +67 more	16 — its uncommon usernames: ecell, guojing, xxaifwq, escheduler, aaden, novinhost, code87, wangyaowei, xjga, inst01 +6 more	373.6K	Germany	DigitalOcean, LLC
1	143.244.183.240	ecell, guojing, xxaifwq, escheduler, novinhost, aaden, xjga, wangyaowei, code87, inst01 +67 more	16 — ecell, guojing, xxaifwq, escheduler, novinhost, aaden, xjga, wangyaowei, code87, inst01 +6 more	136.4K	United States	DigitalOcean, LLC
2	24.199.108.32	guojing, escheduler, ecell, xxaifwq, aaden, novinhost, xjga, wangyaowei, code87, inst01 +67 more	16 — guojing, escheduler, ecell, xxaifwq, aaden, novinhost, xjga, wangyaowei, code87, inst01 +6 more	140.5K	United States	DigitalOcean, LLC
3	146.190.113.228	guojing, ecell, xxaifwq, escheduler, novinhost, aaden, xjga, wangyaowei, code87, inst01 +67 more	16 — guojing, ecell, xxaifwq, escheduler, novinhost, aaden, xjga, wangyaowei, code87, inst01 +6 more	155.4K	United States	DigitalOcean, LLC
4	134.122.62.125	ecell, escheduler, guojing, xxaifwq, novinhost, wangyaowei, aaden, xjga, code87, inst01 +67 more	16 — ecell, escheduler, guojing, xxaifwq, novinhost, wangyaowei, aaden, xjga, code87, inst01 +6 more	159.1K	Netherlands	DigitalOcean, LLC

Figure 12: IPs sharing the username list of 68.183.213.225

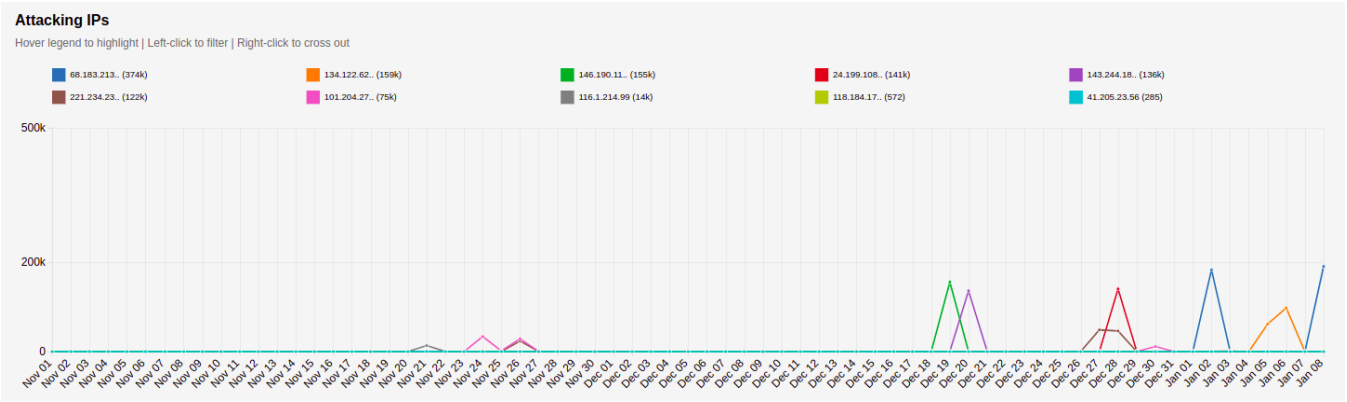


Figure 13: The same IPs in the IP linechart

VI Summary

A Limitations

B Future Work

VII Acknowledgments

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